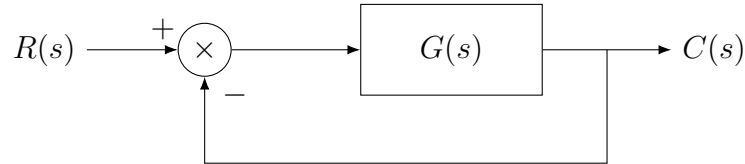


Q1. Find the steady-state error for the system of Figure if

$$G(s) = \frac{5}{s^2 + 7s + 10}$$

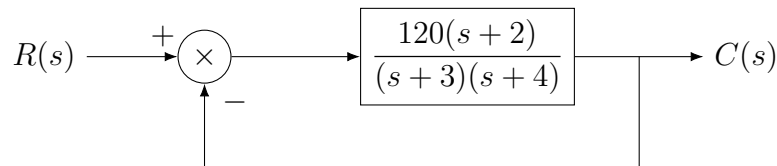
and the input is a unit step.



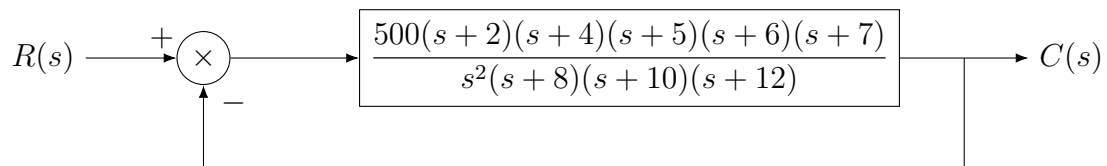
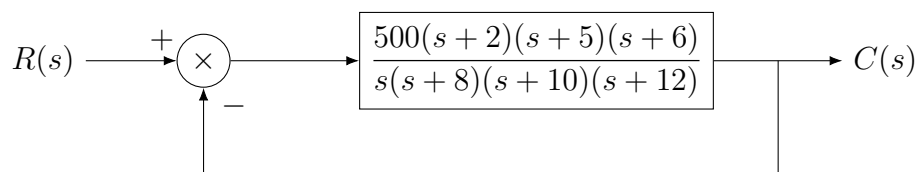
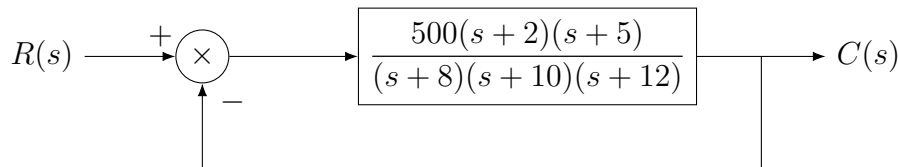
Q2. Find the steady-state errors for the inputs

$$5u(t), \quad 5tu(t), \quad \text{and} \quad 5t^2u(t)$$

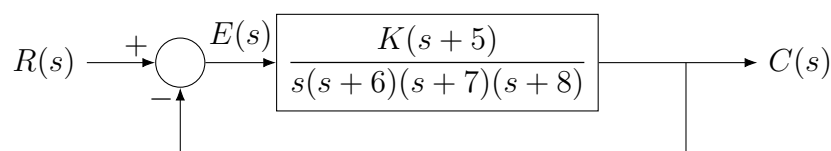
to the system shown in Figure. The function $u(t)$ is the unit step.



Q3. For each system of Figure, evaluate the static error constants and find the expected error for the standard step, ramp, and parabolic inputs.



Q4. Given the control system in Figure, find the value of K so that there is 10% error in the steady state.

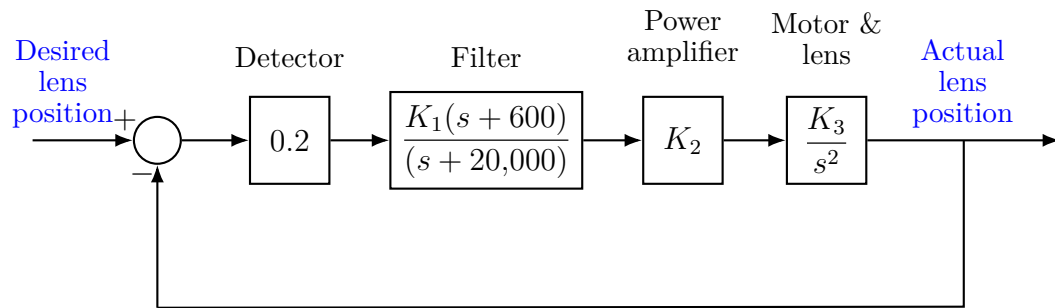


Q5. Evaluate the steady-state error for the system described by

$$A = \begin{bmatrix} -5 & 1 & 0 \\ 0 & -2 & 1 \\ 20 & -10 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 1 & 0 \end{bmatrix}.$$

for unit step and unit ramp inputs.

Q6. Given the video laser disc recording system whose block diagram of the focusing lens position control system is shown in Figure.



- (i) If the focusing lens needs to be positioned to an accuracy of $\pm 0.005 \mu\text{m}$, find the value of $K_1 K_2 K_3$ if the warp on the disc yields a worst-case disturbance in the focus of $15t^2 \mu\text{m}$.
- (ii) Use the Routh–Hurwitz criterion to show that the system is stable when the conditions of part (a) are met.